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Project Title:	Negative Impacts of Altered Gravity Models on Male Mammalian Reproductive Capacity		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	Due to premature termination of funding this grant technical staff that were supported by the grant have departed from the lab.		
COI Name (Institution):	Ronca, April Ph.D. (Wake Forest University School of Medicine) Wade, Charles (US Army Institute of Surgical Research)		
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Task Description:

The overall hypothesis driving the proposed research is that extended exposure to microgravity (μG) has a negative impact on the capacity of males of mammalian species to reproduce. This hypothesis is based on previous published work from our laboratory, demonstrating that hind limb sus-pension (HLS) of adult male rats for 6 weeks causes a block of spermatogenesis leading to permanent sterility. Furthermore, the initial 6 week HLS treatment triggers a decline in testicular function that continues beyond the initial HLS treatment period resulting in massive apoptosis of the germinal epithelium and sterility when assessed 6 mos later. One of the initial insults to the testis caused by HLS is a reduction in blood flow to levels just above the minimum known to sustain spermatogenesis. Essential components of this productive experimental approach have been to utilize i) partial inguinal to prevent the testes from becoming abdominal during hind limb elevation, and ii) pair-fed control animals identical to the HLS animals with inguinal ligation and tail harnessing, but elevated only to lift the tail but not limit contact of the hind limbs with the cage floor.

New goals raised by the results during the previous project period are 1. Determine the maximum duration of HLS that permits subsequent restoration of spermatogenesis and fertility. 2. Ascertain the role that reactive oxygen species (ROS) play in the mechanisms that block spermatogenesis in long term HLS, and whether dietary administration of inhibitors of ROS damage serve as a countermeasure to the loss of spermatogenesis 3. Determine if prepubertal testicular development and adult testicular function are compromised in hypergravity (HG).

Two of the Bioastronautics Roadmap (BR) risks are highlighted by our research:

• Risk 28: Increased cancer morbidity or mortality risk in astronauts may be caused by occupational radiation exposure – May be influenced by other space flight factors including microgravity & environmental contaminants. Mars mission not feasible unless improved shielding is developed. – Exposure to space radiation increases the risk of developing cancer later in life.

• Risk 30: Chronic and Degenerative Tissue Risks – Radiation exposure may result in degenerative tissue diseases. May be caused by occupational radiation exposure or the combined effects of radiation & other space flight factors – Acute or chronic illness due to tissue degeneration may lead to mission impacts, or adverse health consequences after return.

A key question raised by our results is whether the long term hindlimb unloaded model is predictive of the effect of long term microgravity on male reproductive health. This health concern is a separate but related issue to the increased risk of gonadal cancer due to radiation exposure during space flight. It should be noted that there is extensive literature demonstrating that the risk of gonadal cancer is increased by hyperthermia in the cryptorchid model. It is difficult to ascertain whether the time interval between testicular hyperthermia and the occurrence of testicular cancer would be within the time frame of a 30 month mission to Mars. In this regard, the onset of testicular cancers is postulated to be triggered in the post-cryptorchid male after puberty because of the influence of the hormonal milieu. Thus in the adult male with chronic testicular hyperthermia, this interval could be decreased because the adult male hormonal milieu is already present. When combined with increased radiation exposure, this will significantly elevate the risk for testicular cancer (at least 4-10 fold based on the cryptorchid model). In any case, because of the lack of data or investigations focused on the conditions experienced in space flight, these questions warrant more detailed investigation under controlled conditions.

This concern raised by our studies is relevant for human male astronauts in the context of the Bioastronautics Critical Roadmap as well as for rodents as an experimental model for humans. Unfortunately, no animal data is available because the 6 weeks hindlimb unloading is yet to be paralleled by space flight experiments of at least equivalent length. Human data from male astronauts or cosmonauts exposed to Microgravity for longer than Space Shuttle mission duration has not been published or been made available in response to gather such information. Thus it is impossible to ascertain whether or not similar problems have been reported in humans exposed to long term microgravity conditions.

A key question raised by these studies is whether the hindlimb unloaded model is relevant as a model for microgravity in males with scrotal testes. The regulation of testicular temperature well below body temperature is critical for the maintenance of spermatogenesis. The testicular temperature is regulated by a combination of blood flow and position relative to the body. Two opposing forces regulate the position of the testis: 1) the gravitational vector pulls the testicular mass away from the body promoting testicular cooling; 2) the cremasteric muscles pull the testes towards the body thereby increasing testicular temperature. This mode of temperature regulation is identical in all scrotal males including but not limited to humans, primates and rodents. Absence of the gravitational pull on the testis in microgravity eliminates the opposing affect of the cremasteric muscles causing the testes to remain elevated near the body; this is mimicked in the hindlimb unloaded animal because the head downward tilt position prevents gravity from pulling the testis away from the body. Thus microgravity and hindlimb unloading both prevent the testes from cooling properly, causing testicular hyperthermia and compromised spermatogenesis.

Thus, we hypothesize that in long term space flight, chronic testicular hyperthermia may occur in both rats and humans. In relation to the BR, this can cause significantly increased health risks not only for male infertility or sterility, but also for increased testicular cancer due to the well known combined effect of testicular hyperthermia and radiation exposure. Ultimately, this model must be verified/tested by long term rodent and human space flight experiments, developing appropriate countermeasures, and monitoring male astronaut health post flight.

We are thus faced with the following biomedical question: what level of post-flight astronaut cancer and/or infertility risk is acceptable? These risks will magnify with the lengthening duration of space flight and exposure to radiation outside of low earth orbit. The problem of gonadal cancer could even occur within the time frame of a long duration mission such as the 2 year Mars expedition, and post-flight following missions of 6 months or longer on ISS or more likely to the moon. Sterility and subfertility would be manifested post flight in such cases. Is there a willingness of NASA to address these questions by animal and human experimentation, and to address countermeasure development and assessment?

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Male infertility represents approximately half of the cases of infertility in couples. Inflammatory responses to hyperthermia and/or altered blood flow are a common component of male infertility of a broad spectrum of etiologies. The long term hindlimb unloading model appears to reproduce many of the same alterations in testis histopathology associated with many male infertility scenarios. Our studies in relation tp specific aim 3 in particular will study the efficicacy of potential countermeasures for inflammatory and post-iscemic induced male infertity regardless of whether in microgravity or in earth normal gravity.
Task Progress:	During the no cost extension we have continued data analysis of 24ft centrifuge experiments at NASA Ames Research Center and have submitted a paper for publication. The paper's initial review will require some modification for resubmission. Artificial gravity (AG) using centrifugal forces is being explored as a countermeasure for microgravity-induced physiological risks during prolonged space flight. AG research provides the opportunity to investigate consequences of altered G force exposure and potential implications on astronaut health. To determine the effects of long term AG on testicular function, adult male rats were exposed to continuous AG at 2G for 6 weeks and compared to three control groups: stationary (SC), vivarium (VIV), and rotational (RC). After six weeks, half the animals were euthanized and their sperm processed for computer assisted sperm analysis. The remaining animals were subjected to mating trials and tissue analysis. Testes were harvested, weighed, and processed for histology, and blood was collected for hormone analysis. Immediately after 6 week, the AG group relative testes weights were significantly greater than the SC and RC groups. In mating trials, the AG group yielded significantly more pups per litter than the SC group. The AG group also had significantly higher corticosterone levels and significantly lower testosterone levels than the SC and VIV groups, respectively. The data indicate AG at 2G had no detrimental effect on testicular function, consistent with the concept of a biological response along a gravity continuum.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Articles in Peer-reviewed Journals	Timmerberg BD, Wolfe S, Baer LA, Tash JS. "Long term (6 week) artificial gravity at 2G does not impair reproductive health of adult male rats." Journal of Gravitational Physiology, submitted January 2008. , Jan-2008